MapReduce

Big Data Analytics

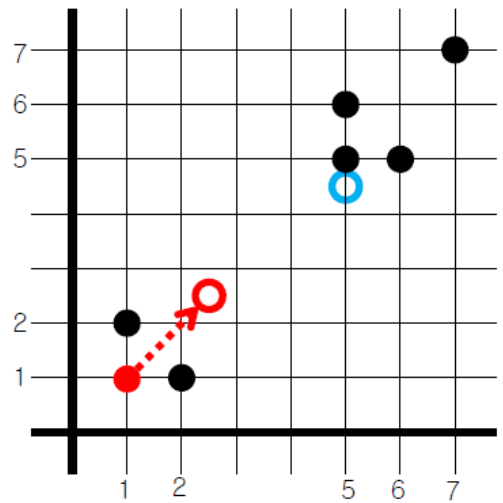
Office hours

Monday 11:30 – 12:30

Thursday 5:00 – 6:00

At 2 402 building A

An Illustration of K-Means Clustering (K = 2)



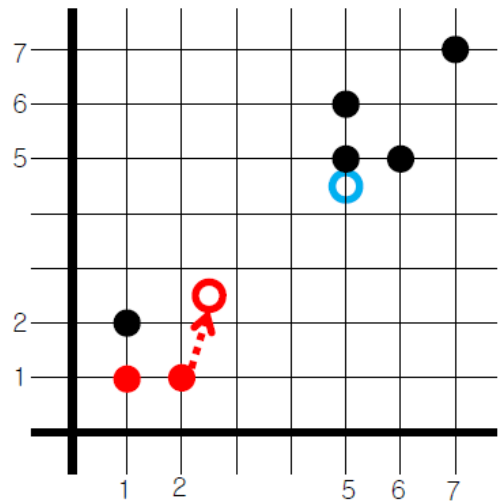
(1, 1) (2, 1) (1, 2) (5, 5) (6, 5) (5, 6) (7, 7)

Class A	Class B
(1, 1)	

Center : (7/3, 8/3) Center : (20/4, 19/4)

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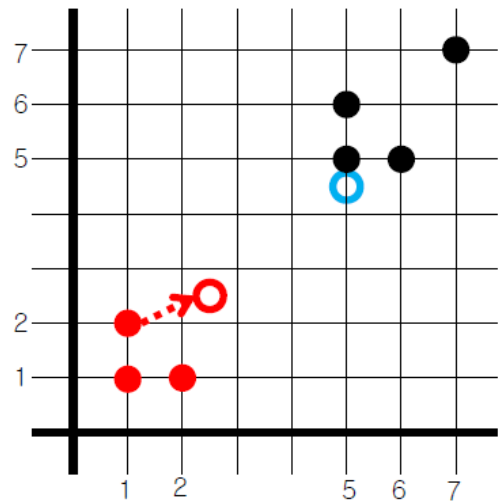
(1, 1) *(2, 1)* (1, 2) (5, 5) (6, 5) (5, 6) (7, 7)

Class A	Class B
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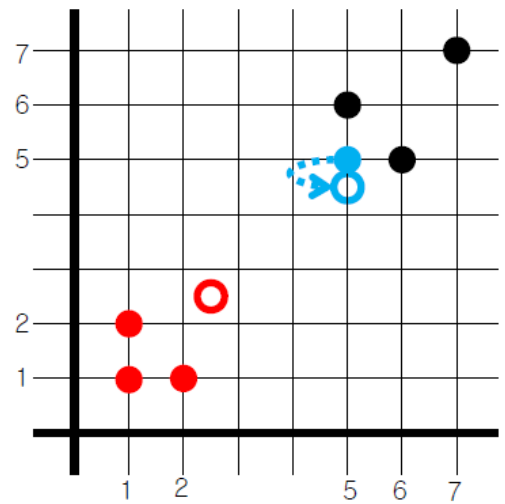
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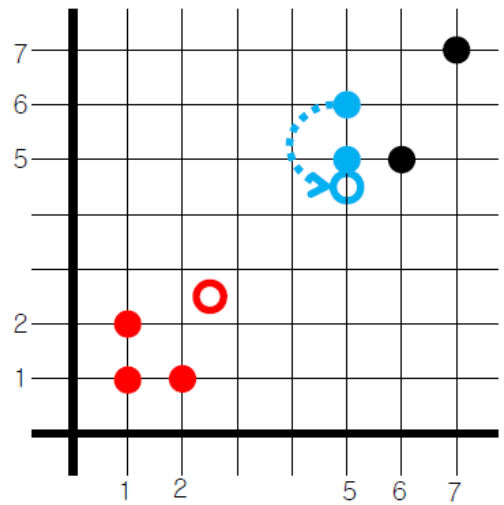
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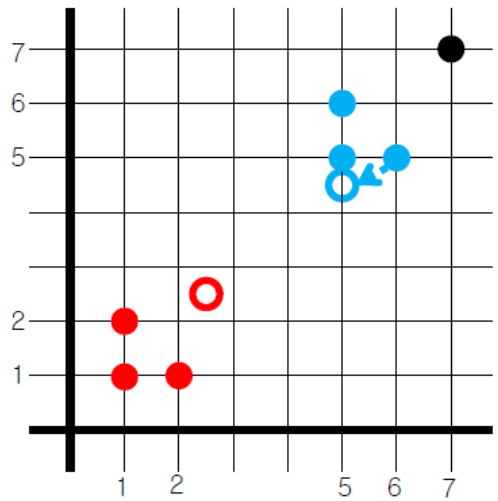
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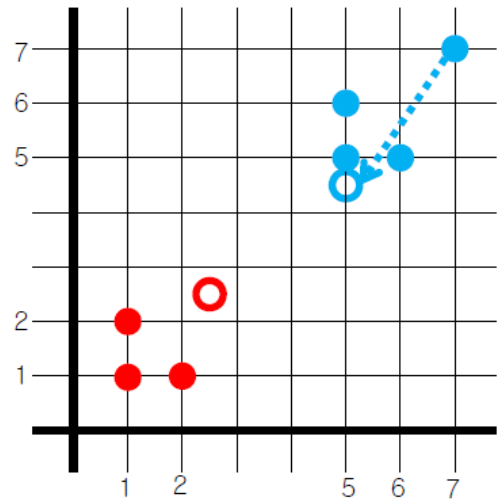
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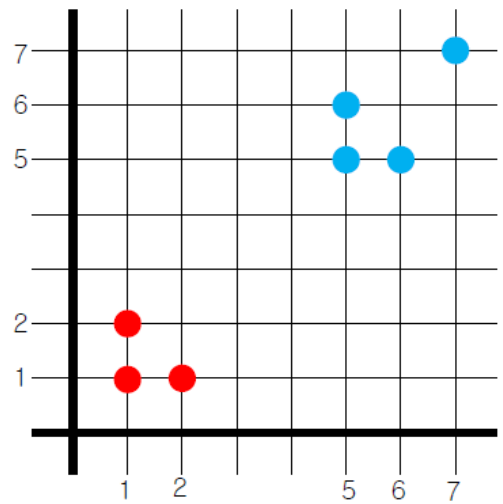
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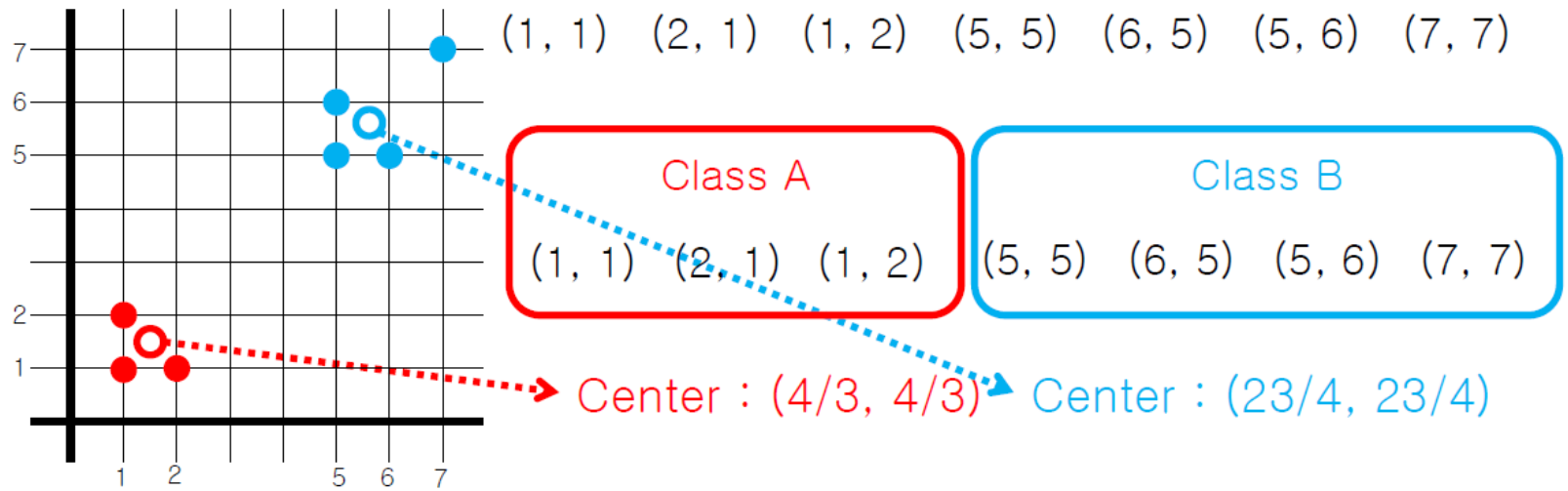
(1, 1) (2, 1) (1, 2)

Class B

(5, 5) (6, 5) (5, 6) (7, 7)

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An Illustration of K-Means Clustering (K = 2)



Update the cluster means

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K-Means Clustering using MapReduce

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K-Means using Map/Reduce

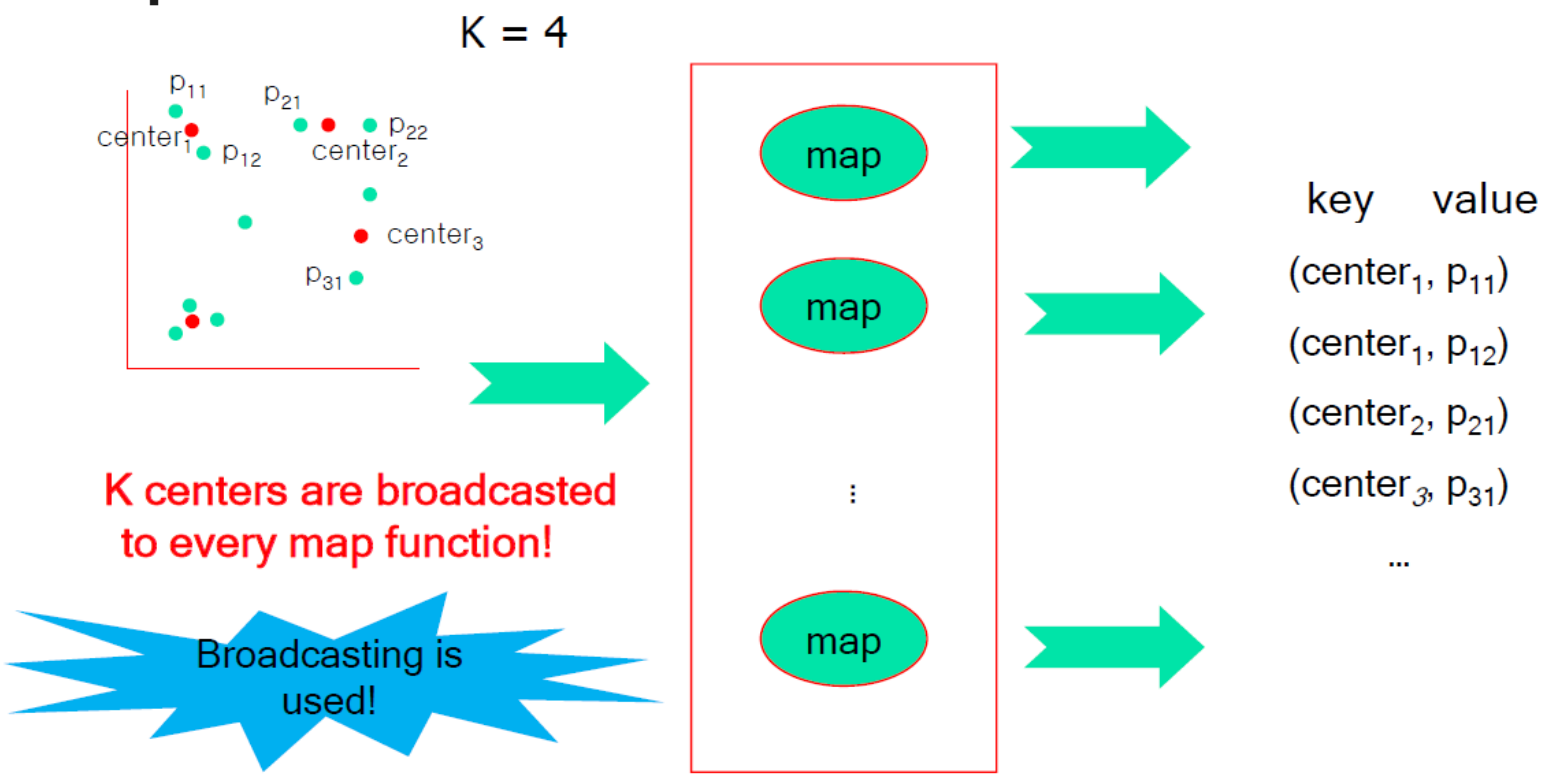
- Iteratively improves partitioning of data into *k* clusters
- Do
  - Map
    - Input is a data point and k centers are broadcasted
    - Finds the closest center among k centers for the input point
  - Reduce
    - Input is one of k centers and all data points having this center as their closest center
    - Calculates the new center using data points
- until all of new centers are not changed

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Performance Metrics of Clustering

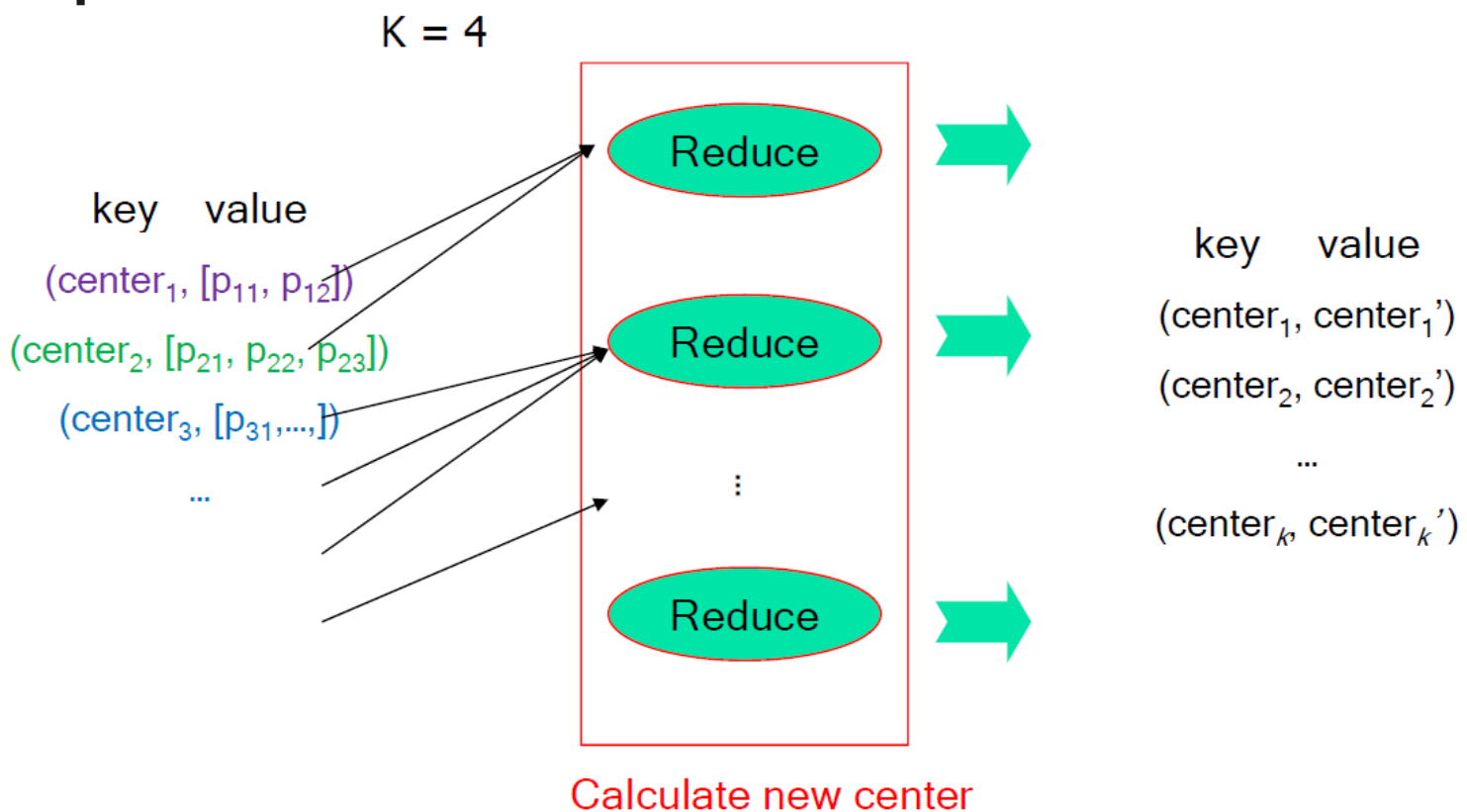
- Silhouette Score
  - (n – i) / max (n, i)
  - n: mean nearest-cluster distance
  - i: mean intra-cluster distance

An Illustration of K-means Clustering: Map



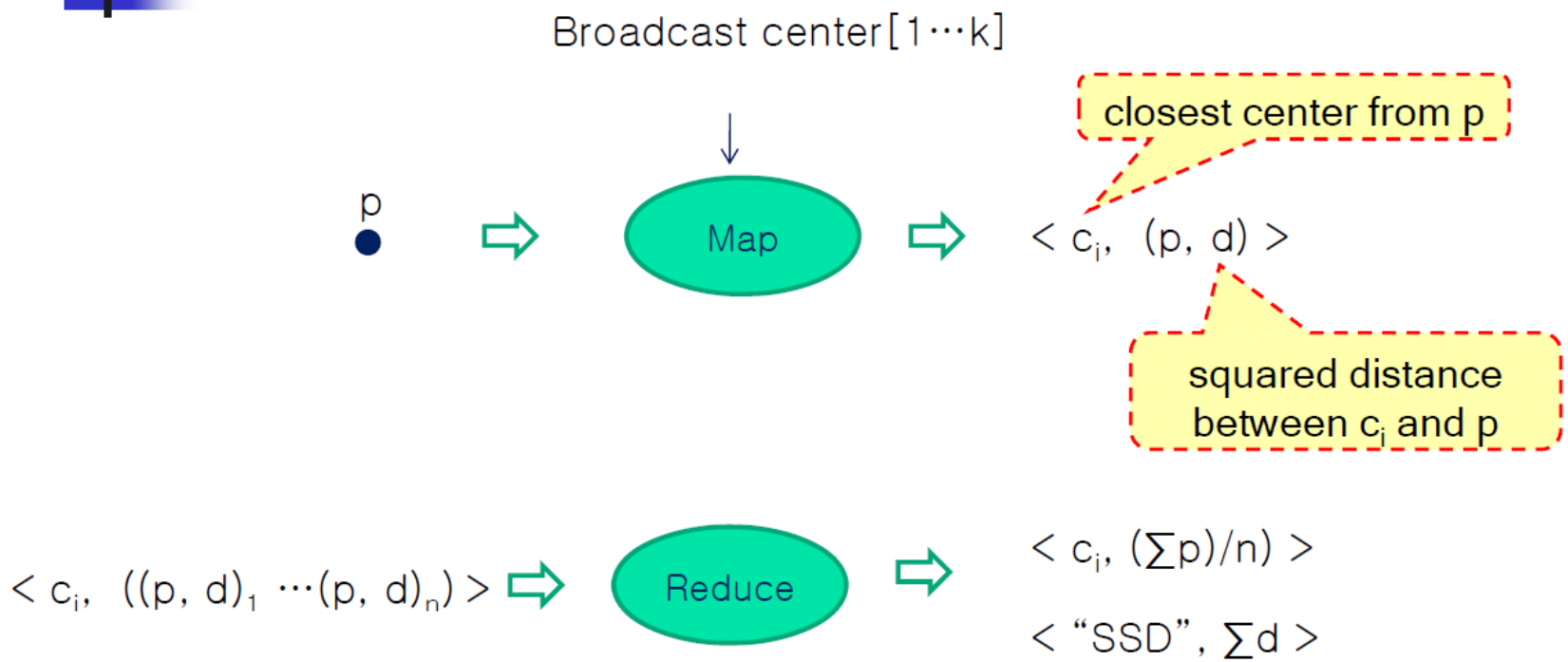
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An Illustration of K-means Clustering: Reduce



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K-means for Large Data



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This Class

- Introduction
  - Hadoop
    - Keys and Values
  - Word count, Page Rank, K-means clustering
- Basic Concepts of Analytics
- Frequent Patterns
- Classification
- Neural Network
- Real world projects in telco business

Data Mining

Big Data Analytics

Readings

- “Data Mining – Concepts and Techniques” by Jiawei Han, Micheline Kamber, Jian Pei, third edition
  - Chap. 1

Data Mining: On What Kinds of Data?

- Database-oriented data sets and applications
  - Relational database, transactional database
- Advanced data sets and advanced applications

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  - Time-series data, sequence data (incl. bio-sequences)

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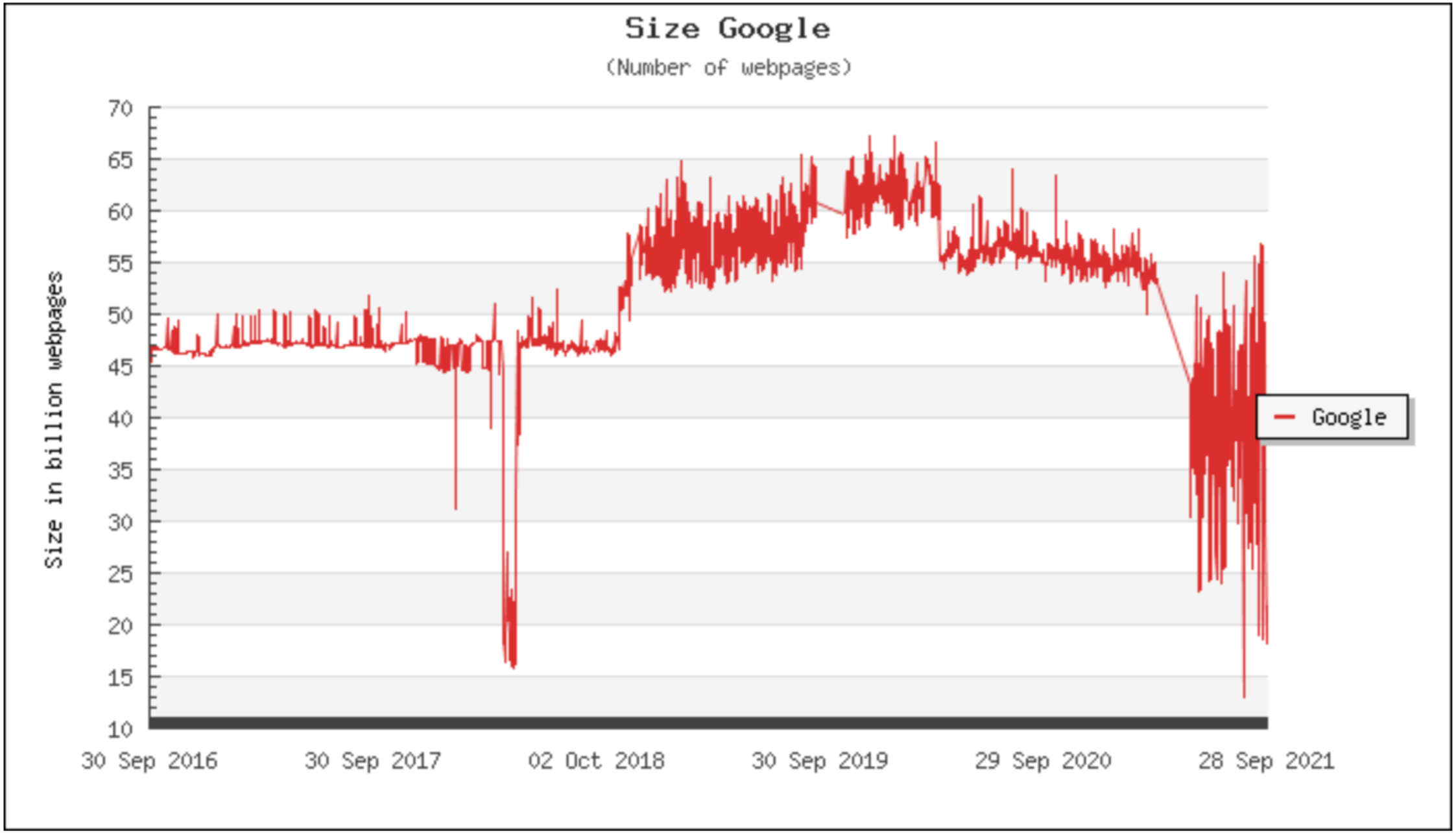
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  - Graphs and social networks data

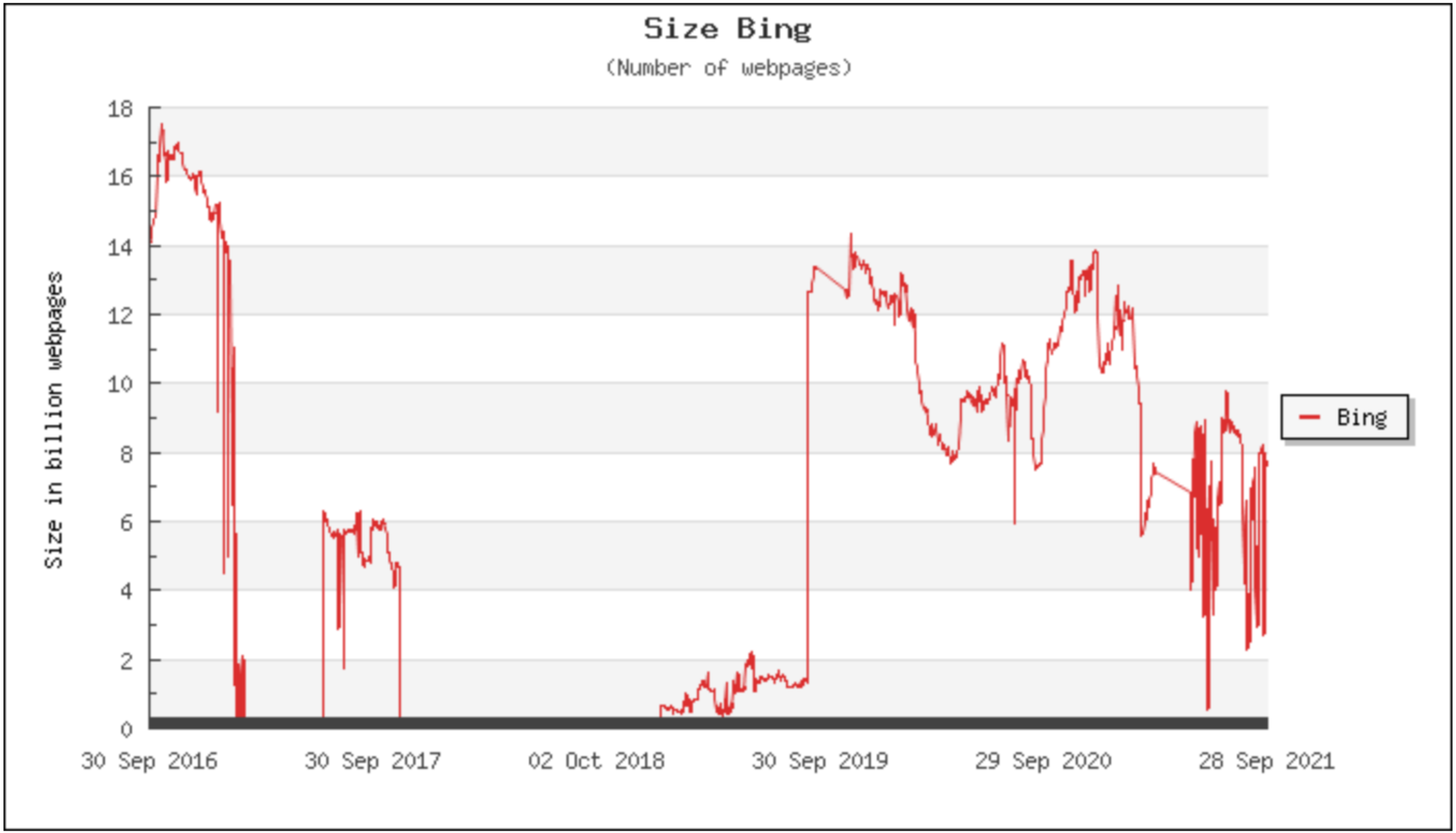
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  - Graphs and social networks data
  - Spatial data
  - The World-Wide Web


Size Google
(Number of webpages)
Size in billion webpages
70
65
60
55
50
45
40
35
30
25
20
15
10
30 Sep 2016
30 Sep 2017
02 Oct 2018
30 Sep 2019
29 Sep 2020
28 Sep 2021
Google


Size Bing
(Number of webpages)
Size in billion webpages
18
16
14
12
10
8
6
4
2
0
30 Sep 2016
30 Sep 2017
02 Oct 2018
30 Sep 2019
29 Sep 2020
28 Sep 2021
Bing

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  - The World-Wide Web
  - Multimedia data

9. 720,000 Hours of Video Uploaded to YouTube Daily



Data Mining: On What Kinds of Data?

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Data Mining Function: Association and Correlation Analysis

- Frequent patterns (or frequent itemsets)
  - What items are frequently purchased together in your Walmart?
- Association, correlation vs. causality
  - A typical association rule
    - Diaper → Beer [0.5%, 75%] (support, confidence)
- How to mine such patterns and rules efficiently in large datasets?

Data Mining Function: Classification

- Classification and label prediction
  - Construct models (functions) based on some training examples
  - Describe and distinguish classes or concepts for future prediction
    - E.g., classify countries based on (climate), or classify cars based on (gas mileage)
  - Predict some unknown class labels
- Typical methods
  - Decision trees, naïve Bayesian classification, support vector machines, neural networks, rule-based classification, pattern-based classification, logistic regression, …
- Typical applications:
  - Credit card fraud detection, direct marketing, classifying stars, diseases, web-pages, …

Data Mining Function: Cluster Analysis

- Unsupervised learning (i.e., Class label is unknown)
- Group data to form new categories (i.e., clusters), e.g., cluster houses to find distribution patterns
- Principle: Maximizing intra-class similarity & minimizing interclass similarity
- Many methods and applications

Data Mining Function: Outlier Analysis

- Outlier analysis
  - Outlier: A data object that does not comply with the general behavior of the data
  - Noise or exception? — One person's garbage could be another person's treasure
  - Methods: by product of clustering or regression analysis, …
  - Useful in fraud detection, rare events analysis

Structure and Network Analysis

- Graph mining
  - Finding frequent subgraphs (e.g., chemical compounds), trees (XML), substructures (web fragments)
- Information network analysis
  - Social networks: actors (objects, nodes) and relationships (edges)
    - e.g., author networks in CS, terrorist networks
  - Multiple heterogeneous networks
    - A person could be multiple information networks: friends, family, classmates, …
  - Links carry a lot of semantic information: Link mining
- Web mining
  - Web is a big information network: from PageRank to Google
  - Analysis of Web information networks
    - Web community discovery, opinion mining, usage mining, …

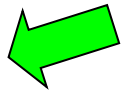
Frequent Patterns

Big Data Analytics

Readings

- “Data Mining – Concepts and Techniques” by Jiawei Han, Micheline Kamber, Jian Pei, third edition
  - Chap. 6

Chapter 6: Mining Frequent Patterns, Association and Correlations: Basic Concepts and Methods

- Basic Concepts 
- Frequent Itemset Mining Methods
- Which Patterns Are Interesting?—Pattern Evaluation Methods
- Summary

What Is Frequent Pattern Analysis?

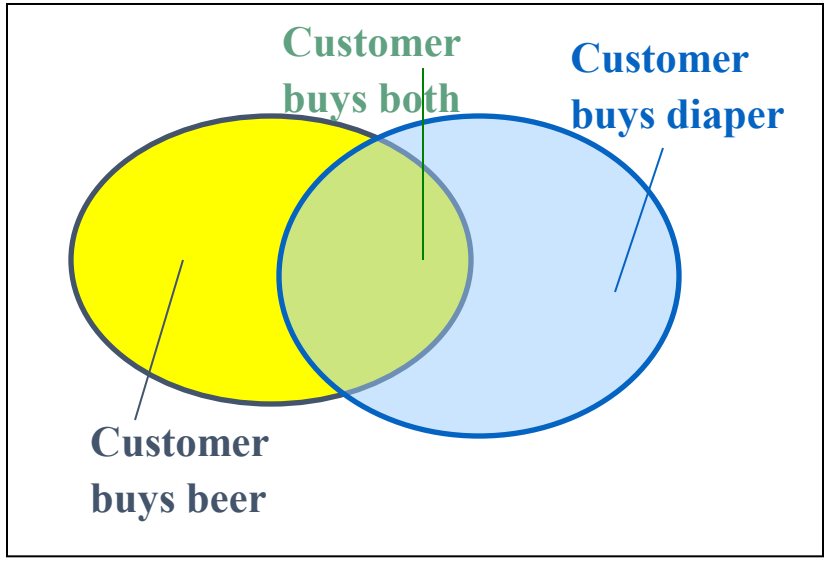
- Frequent pattern: a pattern (a set of items, subsequences, substructures, etc.) that occurs frequently in a data set
- First proposed by Agrawal, Imielinski, and Swami [AIS93] in the context of frequent itemsets and association rule mining
- Motivation: Finding inherent regularities in data
  - What products were often purchased together?— Beer and diapers?!
  - What are the subsequent purchases after buying a PC?
- Applications
  - Basket data analysis, Web log (click stream) analysis, and DNA sequence analysis.

Why Is Freq. Pattern Mining Important?

- Freq. pattern: An intrinsic and important property of datasets
- Foundation for many essential data mining tasks
  - Association, and correlation analysis
  - Sequential, structural (e.g., sub-graph) patterns
  - Cluster analysis: frequent pattern-based clustering

Basic Concepts: Frequent Patterns

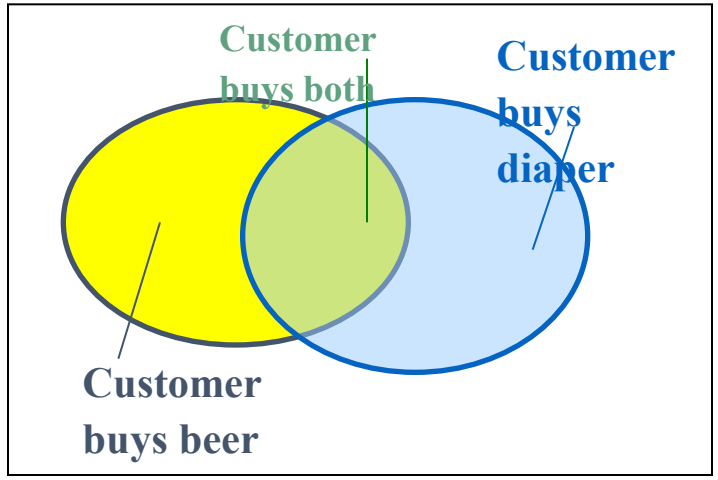
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20	Beer, Coffee, Diaper
30	Beer, Diaper, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk



- itemset: A set of one or more items
- k-itemset $X = \{x_1, \ldots, x_k\}$
- *(absolute) support*, or, *support count* of X: Frequency or occurrence of an itemset X
- *(relative) support*, *s*, is the fraction of transactions that contains X (i.e., the probability that a transaction contains X)
- An itemset X is *frequent* if X's support is no less than a *minsup* threshold

Basic Concepts: Association Rules

Tid	Items bought
10	Beer, Nuts, Diaper
20	Beer, Coffee, Diaper
30	Beer, Diaper, Eggs
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50	Nuts, Coffee, Diaper, Eggs, Milk



- Find all the rules $X \rightarrow Y$ with minimum support and confidence
  - support, *s*, probability that a transaction contains $X \cup Y$
  - confidence, *c*, conditional probability that a transaction having X also contains *Y*

Let minsup = 50%, minconf = 50%

Freq. Pat.: Beer:3, Nuts:3, Diaper:4, Eggs:3, {Beer, Diaper}:3

- Association rules: (many more!)
  - *Beer* → *Diaper* (60%, 100%)
  - *Diaper* → *Beer* (60%, 75%)

Closed Patterns and Max-Patterns

- A long pattern contains a combinatorial number of sub-patterns, e.g., $\{a_1, \ldots, a_{100}\}$ contains $\binom{100}{1} + \binom{100}{2} + \ldots + \binom{1}{1}\binom{0}{0}\binom{0}{0} = 2^{100} - 1 = 1.27*10^{30}$ sub-patterns!

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- An itemset X is closed if X is *frequent* and there exists *no super-pattern* $Y \supset X$, *with the same support* as X (proposed by Pasquier, et al. @ ICDT'99)

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- An itemset X is a max-pattern if X is frequent and there exists no frequent super-pattern $Y \supset X$ (proposed by Bayardo @ SIGMOD'98)

Closed Patterns and Max-Patterns

- Exercise. DB = {<a_1, …, a_{100}>, < a_1, …, a_{50}>}
  - Min_sup = 1.
- What is the set of closed itemset?

Closed Patterns and Max-Patterns

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- What is the set of closed itemset?
  - <a_1, …, a_{100}>: 1
  - < a_1, …, a_{50}>: 2
- What is the set of max-pattern?

Closed Patterns and Max-Patterns

- Exercise. DB = {<a_1, …, a_{100}>, < a_1, …, a_{50}>}
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  - < a_1, …, a_{50}>: 2
- What is the set of max-pattern?
  - <a_1, …, a_{100}>: 1

Computational Complexity of Frequent Itemset Mining

- How many itemsets are potentially to be generated in the worst case?
  - The number of frequent itemsets to be generated is senstive to the minsup threshold
  - When minsup is low, there exist potentially an exponential number of frequent itemsets

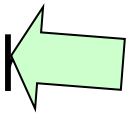
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- The worst case complexty vs. the expected probability
  - Ex. Suppose Walmart has 10^4 kinds of products
    - The chance to pick up one product 10^{-4}
    - The chance to pick up a particular set of 10 products: ~10^{-40}

Scalable Frequent Itemset Mining Methods

- Apriori: A Candidate Generation-and-Test Approach 
- Improving the Efficiency of Apriori
- FPGrowth: A Frequent Pattern-Growth Approach
- ECLAT: Frequent Pattern Mining with Vertical Data Format

The Downward Closure Property and Scalable Mining Methods - Apriori

- The downward closure property of frequent patterns
  - Any subset of a frequent itemset must be frequent
  - If **{beer, diaper, nuts}** is frequent, so is **{beer, diaper}**
  - i.e., every transaction having {beer, diaper, nuts} also contains {beer, diaper}

TID	Items
1	Beer, Diaper, Eggs
2	Beer, Diaper, Nuts
3	Diaper, Nuts, Eggs, Coffee
4	Beer, Diaper, Nuts, Eggs, Coffee, Milk

Apriori: A Candidate Generation & Test Approach

- Apriori pruning principle: If there is any itemset which is infrequent, its superset should not be generated/tested! (Agrawal & Srikant @VLDB'94, Mannila, et al. @ KDD' 94)
- Method:
  - Initially, scan DB once to get frequent 1-itemset
    - {Beer}, {Nuts}, {Diaper}, {Eggs}

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  - Initially, scan DB once to get frequent 1-itemset
    - {Eggs}
  - Generate length (k+1) candidate itemsets from length k frequent itemsets
    - {Beer, Nuts}, {Beer, Diaper}, {Beer, Eggs}, {Nuts, Diaper}, {Nuts, Eggs}, {Diaper, Eggs}

Apriori: A Candidate Generation & Test Approach

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    - {Beer, Nuts}, {Beer, Diaper}, {Beer, Eggs}, {Nuts, Diaper}, {Nuts, Eggs}, {Diaper, Eggs}
  - Test the candidates against DB

Tid	Items bought
10	Beer, Nuts, Diaper
20	Beer, Coffee, Diaper
30	Beer, Diaper, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk

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- Method:
  - Initially, scan DB once to get frequent 1-itemset
    - {Beer}, {Nuts}, {Diaper}, {Eggs}
  - Generate length (k+1) candidate itemsets from length k frequent itemsets
    - {Beer, Nuts}, {Beer, Diaper}, {Beer, Eggs}, {Nuts, Diaper}, {Nuts, Eggs}, {Diaper, Eggs}
  - Test the candidates against DB
  - Terminate when no frequent or candidate set can be generated

The Apriori Algorithm (Pseudo-Code)

C_k: Candidate itemset of size k
L_k : frequent itemset of size k

L_1 = {frequent items};
for (k = 1; L_k !=∅; k++) **do begin**
    C_{k+1} = candidates generated from L_k;
    for each transaction t in database do
      increment the count of all candidates in C_{k+1} that
      are contained in t
    L_{k+1} = candidates in C_{k+1} with min_support
    end
return $\cup_k L_k$;

Implementation of Apriori

- How to generate candidates?
  - Step 1: self-joining L_k
  - Step 2: pruning
- Example of Candidate-generation
  - L_3={*abc, abd, acd, ace, bcd*}
  - Self-joining: L_3*L_3
    - *abcd* from *abc* and *abd*
    - *acde* from *acd* and *ace*
  - Pruning:
    - *acde* is removed because *ade* is not in L_3
  - C_4 = {*abcd*}

The Apriori Algorithm—An Example with Sup_{min} = 2

Database TDB

Tid	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

C_1

1st scan →

The Apriori Algorithm—An Example with Sup_{min} = 2

Database TDB

Tid	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

C_1

1st scan

Itemset	sup
{A}	2
{B}	3
{C}	3
{D}	1
{E}	3

L_1

The Apriori Algorithm—An Example with Sup_{min} = 2

Database TDB

Tid	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

C_1

1st scan

Itemset	sup
{A}	2
{B}	3
{C}	3
{D}	1
{E}	3

L_1

Itemset	sup
{A}	2
{B}	3
{C}	3
{E}	3

The Apriori Algorithm—An Example with Sup_{min} = 2

Database TDB

Tid	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

C_1

1st scan

Itemset	sup
{A}	2
{B}	3
{C}	3
{D}	1
{E}	3

L_1

Itemset	sup
{A}	2
{B}	3
{C}	3
{E}	3

C_2

Itemset
{A, B}
{A, C}
{A, E}
{B, C}
{B, E}
{C, E}

The Apriori Algorithm—An Example with Sup_{min} = 2

Database TDB

Tid	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

C_1

1st scan

Itemset	sup
{A}	2
{B}	3
{C}	3
{D}	1
{E}	3

L_1

Itemset	sup
{A}	2
{B}	3
{C}	3
{E}	3

C_2

2nd scan

Itemset
{A, B}
{A, C}
{A, E}
{B, C}
{B, E}
{C, E}

The Apriori Algorithm—An Example with Sup_{min} = 2

Database TDB

Tid	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

1st scan

C_1

Itemset	sup
{A}	2
{B}	3
{C}	3
{D}	1
{E}	3

L_1

Itemset	sup
{A}	2
{B}	3
{C}	3
{E}	3

C_2

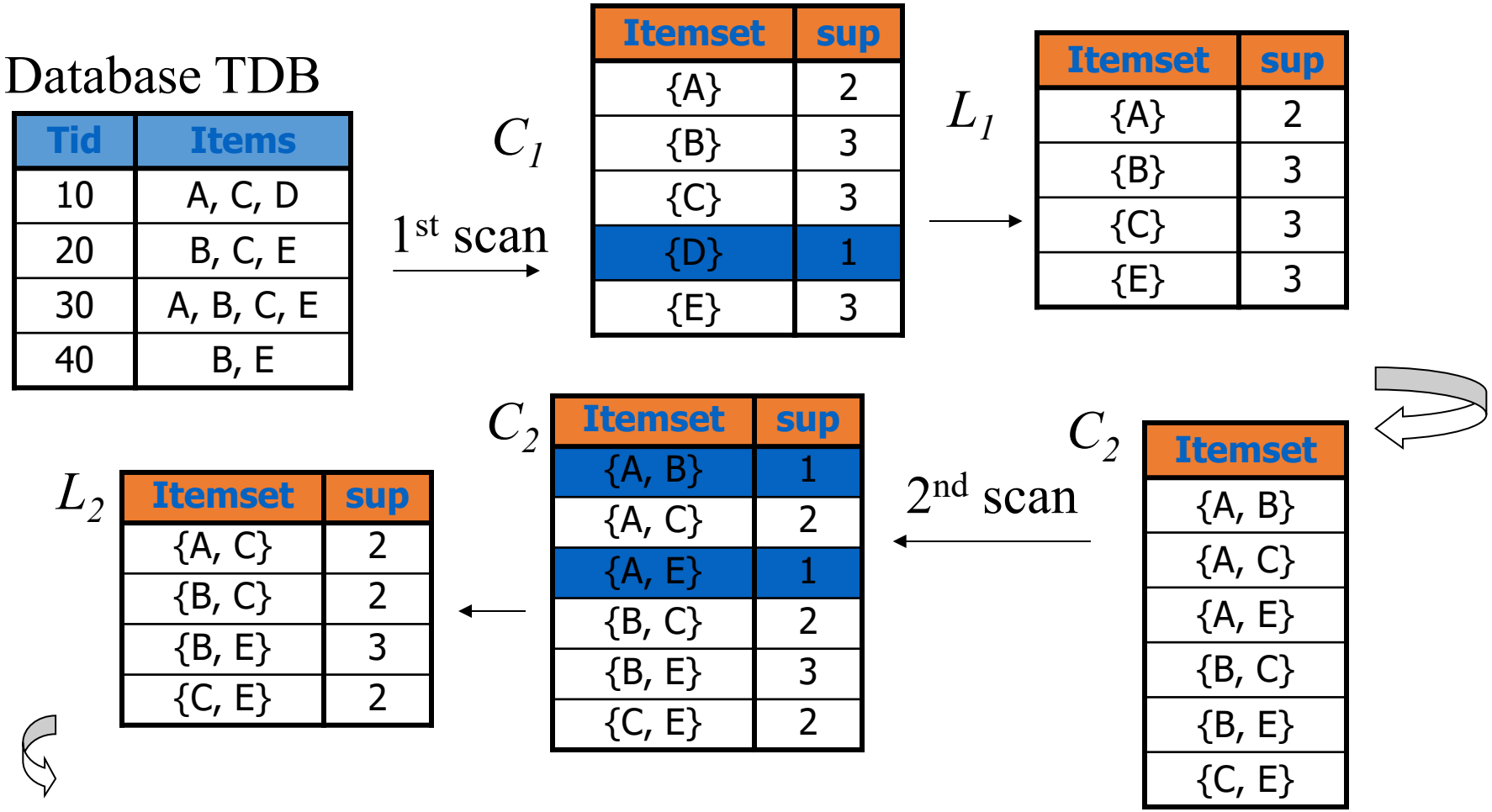
Itemset	sup
{A, B}	1
{A, C}	2
{A, E}	1
{B, C}	2
{B, E}	3
{C, E}	2

2nd scan

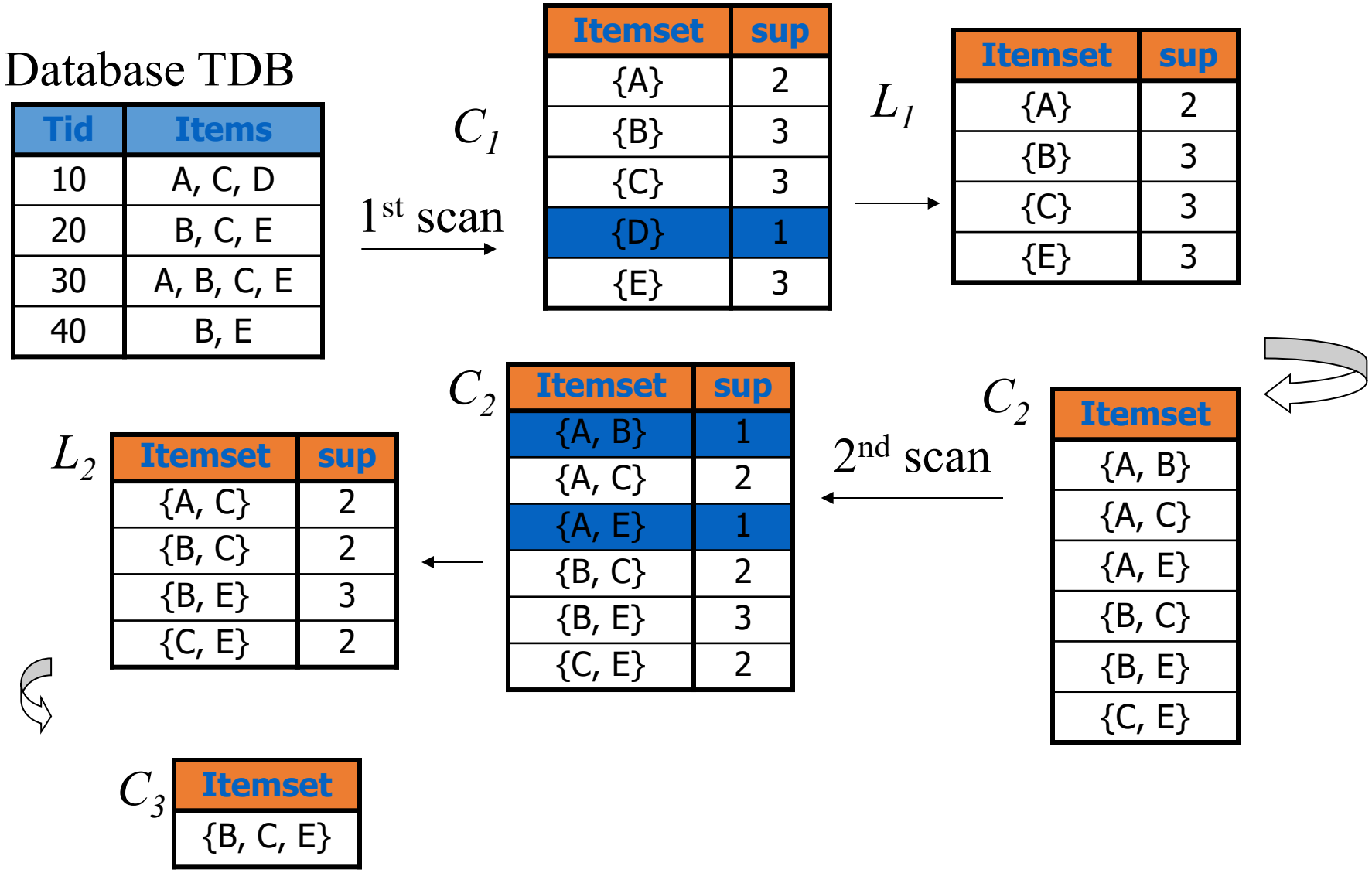
C_2

Itemset
{A, B}
{A, C}
{A, E}
{B, C}
{B, E}
{C, E}

The Apriori Algorithm—An Example with Sup_{min} = 2



The Apriori Algorithm—An Example with Sup_{min} = 2



The Apriori Algorithm—An Example

